

Jurnal Perdagangan

Arsitektur Agen AI

Kerangka Kecerdasan Multi-Model — v1.5.0

Claude Sonnet · Claude Haiku · Google Gemini · xAI Grok · Nansen
CoinGecko · Coinalyze · Finnhub · Deribit · DefiLlama · blockchain.com · yfinance · CCXT

Apa yang dibahas dokumen ini

This document describes every AI agent, data agent, and automation agent that powers the self-hosted crypto futures trading journal. Each section explains what the agent does, why it was designed that way, what model it uses, when it fires, and how it connects to the others. Suitable for beginners and experts.

1. Gambaran Umum Sistem

The trading journal is a self-hosted application running on a Raspberry Pi 5. It connects to cryptocurrency futures exchanges (Bitget, Blofin), tracks your trade history, and uses a network of specialised AI agents to help you make better trading decisions. Every agent has a specific, focused responsibility — none of them tries to do everything.

For **beginners**: think of it as a team of analysts, each an expert in one area. One analyst reads the charts, another checks social media, a third scores your trade ideas, another reviews your history. The orchestrator is the team leader who decides who speaks and how much weight to give each opinion.

For **experts**: the architecture uses a prompt-caching-aware stable/dynamic context split, a consensus scoring layer across two independent LLMs with divergence detection, MC-weighted external intelligence (Grok), and a backtest feedback loop that injects historical win-rate patterns directly into every scoring prompt.

Tiga jenis agen

Jenis	Artinya	Contoh
Agen Analisis	Receive input (a trade call, a position, historical data) and analyze, assess, and provide a trade assessment.	data_analysis, assistant, trade_assessment
Agen Otomatisasi	Run on a schedule in the background without any user interaction. They fetch data, detect setups, send alerts.	scan_scheduler, data_fetcher, detector, setup_sender
Agen Data	Fetch and cache external data (no AI). They are called by analysis agents to enrich prompts with external data.	crypto_prices, news_feed, market_data

2. Orkestrator Utama

The orchestrator (`agent_orchestrator.py`) does not call any AI model itself. Its job is to coordinate results from multiple models and make routing decisions. It answers two questions: *which model should handle this task?* and *do Claude and Gemini agree on this trade signal?*

2.1 Pengarah Model

Every AI task in the journal is classified as either a **reasoning task** (needs the most capable model) or a **classification task** (can be done by a faster, cheaper model). This matters because Claude Sonnet costs roughly 10× more per token than Claude Haiku — routing correctly reduces costs without sacrificing accuracy.

Tugas	Model	Alasan pemilihan
call_analyzer	Sonnet 4.6	Keluaran JSON terstruktur kompleks 15+ bidang, penilaian chain-of-thought
scanner_batch	Sonnet 4.6	Mengevaluasi 12 simbol sekaligus, butuh penalaran masuk/SL/TP mendalam
advisor	Sonnet 4.6	Pelatihan portofolio penuh dari 800+ perdagangan, rekomendasi panjang
rulebook	Sonnet 4.6	Menyintesis seluruh riwayat perdagangan menjadi aturan perdagangan personal
limit_analyzer	Sonnet 4.6	Keputusan risiko pada order tertunda — akurasi sangat penting
pattern_detector	Sonnet 4.6	Analisis penggabungan lintas-pola — membutuhkan penalaran nyata
scanner_quick	Haiku 4.5	Skor 0-10 + satu kalimat — klasifikasi murni, berjalan 100× per pemindaian
live_trade	Haiku 4.5	Tindakan Tahan/Tutup/Sesuaikan — rubrik sederhana, latensi penting
hindsight	Haiku 4.5	Vonis MASUK/LEWATI retroaktif — tugas klasifikasi biner
trade_grader	Haiku 4.5	Nilai eksekusi A/B/C/D — rubrik sederhana, berjalan sekali per perdagangan ditutup

2.2 Algoritma Penilaian Konsensus

When a user analyzes a trade call, both Claude and Google Gemini score the setup independently. Claude receives the full context (rulebook, chart data, market conditions, similar historical trades). Gemini receives *only the raw call text* — no extra context. This is intentional: two assessors with different information sets produce more meaningful agreement or disagreement than two copies of the same prompt.

When they agree ($|\Delta| \leq 1$), confidence is high — the setup signal is robust. When they strongly disagree ($|\Delta| > 3$), the trade is flagged for manual review before acting on it. The weighted average (Claude 60%, Gemini 40% on mild disagreements) reflects that Claude's full context generally produces more accurate scores for structured technical setups.

Delta ($ \text{Claude} - \text{Gemini} $)	Kepercayaan	Skor digunakan	Tanda UI	Tindakan disarankan
0 – 1 point	Tinggi	Rata-rata sederhana	✓ Dikonfirmasi	Perdagangan dengan risiko normal
2 points	Sedang	Rata-rata sederhana	~ Selaras	Perdagangan, pantau ketat
3 points	Rendah	Claude 60% + Gemini 40%	⚠ Divergen	Kurangi ukuran atau tunggu konfirmasi
> 3 points	Sangat Rendah	Skor Claude dipakai	■ TINJAU	Jangan perdagangan — selidiki ketidaksetujuan

3. Agen Analisis

Analysis agents are triggered by user actions (clicking a button, requesting an analysis). Each one receives a focused input, builds an optimised prompt from the shared context system, calls the AI, and returns structured JSON.

■ Call Analyzer — ai_call.py

Sonnet 4.6

Trigger: Pengguna menempel teks panggilan analis (Telegram, Twitter, manual)

Output: Skor 1-10, masuk/SL/TP, ukuran posisi, peringatan pola, R:R, konsensus Gemini

Agen kecerdasan utama untuk mengevaluasi panggilan perdagangan dari analis eksternal.

When a user receives a trade call from an analyst on Telegram, they paste it here. The agent automatically extracts the symbol, direction, entry price, stop loss and take profit levels using regex. It then runs three things in parallel: (1) an ATR-based stop loss quality check using 1H candle data, (2) live market context (funding rates, Fear & Greed), and (3) Gemini's independent pre-proof score. Claude then receives the full context — personalised rulebook, calibration feedback, chart indicators for 4H and 1D, Nansen smart money signal, Grok social sentiment (weighted by market cap), similar historical trades — and produces a 15-field structured analysis. The previous analysis for the same symbol is injected as a learning loop (CoT reuse), enabling Claude to detect if setup conditions have changed. The consensus score (Claude vs Gemini) is saved to the database alongside the full analysis.

■ Scanner — ai_scanner.py (3-stage)

Sonnet + Haiku

Trigger: Setiap 30 menit (otomatis) atau dijalankan manual

Output: Hingga 12 setup dengan skor: zona_masuk, sl, tp1, tp2, rasio_rr, urgensi

Secara proaktif menemukan setup perdagangan dari 100+ simbol tanpa menunggu panggilan analis.

Stage 0 (Macro Layer, once per run): VIX, Fear & Greed, Finnhub economic calendar, and BTC dominance are fetched once and stored in scan state. Score caps are computed: VIX > 35 → cap 6.0, VIX 25-35 → cap 7.5, high-impact macro event in 24h → cap 7.0. These caps are applied to every Stage 3 score BEFORE the threshold check — a 9-scoring setup in a VIX > 35 environment is capped to 6.0. Macro context and warnings are visible in the scanner UI. Stage 1 (Confluence Filter, no AI): fetches 4H and 1D candles for all symbols in parallel and computes RSI, MACD, EMA, ADX, WaveTrend, CVD, and 9-signal confluence score including SMT divergence. Symbols below threshold are eliminated — typically cuts 100+ symbols to ~25-30 with zero API cost. Stage 2 (Quality Gate, no AI): applies technical rules — rejects overextended RSI, missing S/R structure, flat ADX, very high funding rate. Cuts to ~10-15. Stage 3a (Haiku Quick Score): Haiku scores each finalist with a minimal prompt (120 tokens output max) — faster and 10× cheaper than Sonnet. Setups scoring below threshold are dropped. Stage 3b (Sonnet Batch + macro cap): all remaining finalists are scored in a SINGLE Sonnet call using a batch prompt. Macro cap is applied to each score before the threshold. This is a key token optimisation — scoring 12 symbols simultaneously rather than 12 sequential calls. Stage 3c (Gemini Consensus): top-5 finalists receive an independent Gemini score. The final ranking adjusts based on consensus confidence. Alerted setups are automatically saved to analyzed_calls so they can be linked to live positions without manual intervention.

■ AI Advisor — ai_advisor.py

Sonnet 4.6

Trigger: Pengguna mengklik 'Dapatkan Saran AI' di Edge Lab

Output: Kekuatan portofolio, kelemahan, rekomendasi spesifik, wawasan simbol

Pelatihan portofolio tingkat tinggi berdasarkan riwayat perdagangan lengkap dan pasar terkini.

The advisor receives aggregated statistics from your entire trade history: win rate, profit factor, performance by symbol, by weekday, by hour, by setup type, by duration. It also receives the current market context (BTC dominance, Fear & Greed, funding rates). The rulebook and calibration data are cached as the stable prefix. Claude identifies patterns across all dimensions and produces a structured coaching report: what you're doing well (strengths), where you're losing money (weaknesses), and 3-5 specific actionable recommendations with data to back them up. For example: 'Your Friday afternoon LONG trades have a 43% win rate vs 71% on other days — consider avoiding new positions after 15:00 UTC on Fridays.'

■ Hindsight Analyzer — ai_hindsight.py

Haiku 4.5

Trigger: Pengguna menjalankan batch tinjauan balik (N perdagangan terakhir)

Output: Vonis MASUK/LEWATI per perdagangan, akurasi TP/FP/TN/FN, perbandingan P&L;

Penilaian buta retroaktif: apa yang akan dikatakan AI jika melihat setup sebelum hasilnya?

This is a backtesting tool with a key discipline: Claude is shown the technical picture as it appeared AT ENTRY TIME — not the current chart. The agent fetches historical OHLCV candles ending at the exact trade entry timestamp and reconstructs the indicator state. It then scores the setup without knowing how the trade ended. The results show: (1) how often Claude's ENTER calls actually won (True Positive rate), (2) how often its SKIP calls would have been correct, and (3) the hypothetical P&L; if you had only taken trades Claude would rate ≥ 6 . This calibration loop helps identify whether the AI's scoring is predictive of actual outcomes — the foundation for the 85% accuracy target.

■ Live Trade Checker — ai_live_trade.py

Haiku 4.5

Trigger: Pengguna mengklik '■ Analisis AI' pada kartu posisi live

Output: Tindakan (Tahan/Tutup/Sesuaikan SL), peringkat risiko 1-10, saran TP/SL

Pemeriksaan kesehatan cepat per posisi untuk posisi futures terbuka.

Receives the full position data (entry, mark, SL, TP, duration, margin, unrealized P&L; funding rate) plus the current 4H chart indicators. Haiku evaluates whether the trade thesis is still valid, whether the stop is at risk, and whether the position has been open too long. Returns a structured recommendation in under 2 seconds. Haiku is used here specifically because the user is looking at their live portfolio and latency matters — a 5-second wait for each card would be frustrating.

■ Limit Analyzer — ai_limit.py

Sonnet 4.6

Trigger: Pengguna mengklik 'Analisis' pada order limit tertunda

Output: Skor kualitas masuk, penilaian risiko, validasi ATR

Mengevaluasi order limit sebelum terpicu — menangkap setup buruk sebelum terisi.

Limit orders represent planned trades that haven't executed yet. The limit analyzer checks whether the planned entry price is at a structurally sound level, whether the stop loss is outside the ATR noise floor, and how this limit fits with other open or pending positions. Uses Sonnet because this is a consequential decision (real money at risk when the limit fills) and the nuanced entry quality assessment benefits from the more capable model.

■ Trade Grader — ai_trade_grader.py

Haiku 4.5

Trigger: Pengguna mengklik '■ Nilai' pada perdagangan yang sudah ditutup

Output: Nilai A/B/C/D dengan penjelasan tertulis tentang kualitas eksekusi

Loop umpan balik kualitas eksekusi — apakah masuk/keluar benar-benar dilaksanakan dengan baik?

Separate from whether the trade was profitable, the grader evaluates *how well* the trade was executed: did the entry happen near the ideal zone, was the stop set correctly relative to structure, was the exit too early or too late, was risk sizing appropriate? A trade can be a B+ execution that still lost money (correct process, bad luck) or a D execution that won by chance. Tracking execution grades over time identifies whether losses come from bad setups or bad execution — very different problems requiring different solutions.

■ Rulebook Generator — ai_rulebook.py

Sonnet 4.6

Trigger: Regenerasi otomatis mingguan (jika 5+ perdagangan baru) atau pembaruan manual

Output: 10 aturan personal dengan tingkat kepercayaan dan anotasi kadaluarsa

Buku aturan perdagangan yang diperbarui sendiri, disintesis dari riwayat perdagangan Anda.

The rulebook is the most impactful agent for long-term accuracy improvement. Claude reads your complete trade statistics — performance by setup type, symbol, session, weekday, hour, holding period, and direction — and synthesises 5-10 personalised rules. These are not generic advice but rules derived from YOUR specific data. For example: 'Friday morning breakouts: 3 trades, 0 wins, avg -\$166 — strong evidence to avoid.' Rules older than 30 days are annotated [stale] so Claude discounts them in analysis prompts. A regen guard prevents regeneration if fewer than 5 new trades exist since the last update — avoiding rules based on too little data. All 10 rules are injected into every analysis prompt as the cached stable prefix, meaning Claude always has your personalised context without paying for it on every call.

4. Penyedia AI Eksternal

Two external AI providers are integrated alongside Claude to provide independent signals that cannot come from technical analysis alone.

4.1 Google Gemini — Pemberi Skor Pra-Konfirmasi Independen

Why Gemini? Having a second AI model score the same trade call independently creates a cross-validation signal. Gemini sees only the raw call text — no rulebook, no chart context. This information asymmetry is deliberate. If two models with different training data and different context reach the same score, the signal is stronger. If they strongly disagree, that disagreement is itself informative.

For beginners: Imagine having two doctors give independent opinions on the same X-ray without telling either what the other said. If both say 'this looks fine,' you're confident. If one says 'this is serious' and the other says 'nothing to worry about,' you know to get a third opinion.

Technical details: Uses Gemini 2.0 Flash (fast, cheap) via the Google Generative Language API with `responseMimeType: application/json` to force structured output. Runs in parallel with ATR checks and market context fetches — no additional wall-clock time. Cached 30 minutes per (symbol, direction) pair. Results stored in `analyzed_calls.gemini_score` and `consensus_score` columns for backtesting.

4.2 xAI Grok — Kecerdasan Sosial (X/Twitter)

Why Grok? Grok has real-time access to X (Twitter) data. For small-cap and micro-cap crypto assets, the price is often driven more by social narrative and community sentiment than by on-chain fundamentals or technical patterns. Grok is the only AI in the stack that can see what people are saying about a specific coin right now.

Market cap weighting: For large-cap coins like Bitcoin or Ethereum, social media noise far exceeds the signal — institutional trading dominates, and a viral tweet rarely moves the price meaningfully. For micro-cap coins (\$200M market cap or below), a single influential post can move price 30%. The weight formula reflects this reality:

Kapitalisasi Pasar	Bobot Grok	Alasan
> \$5 miliar	0% — dilewati	Kapitalisasi besar: kebisingan sosial > sinyal. Aliran institusional mendominasi.
\$1M – \$5M	15% bobot	Menengah-besar: hanya konteks tambahan.
\$200jt – \$1M	40% bobot	Kecil: sentimen sosial adalah pendorong harga yang berarti.
< \$200jt	80% bobot	Mikro: narasi sosial sering menjadi pendorong UTAMA.
Kapitalisasi tidak diketahui	60% bobot	Diperlakukan sebagai kecil hingga CoinGecko mengkonfirmasi.

What Grok provides: a 100-130 word brief covering X/Twitter sentiment (bullish/bearish/mixed and dominant narratives), recent news or developments (last 7 days), social quality assessment (organic analysis vs. coordinated hype), and the biggest social/news risk to the current trade direction. Red flags are explicitly marked with ■. The brief is injected into the prompt context with a label showing the weight so Claude knows how much to rely on it relative to technical indicators.

5. Agen Data

Data agents fetch, cache, and format external data. They don't call any AI model — they are pure data pipelines whose output enriches AI prompts.

All sources feed into a single **CollectorResult** TypedDict via **data_sources.py** (thin adapter layer). Adding a new source = one function in `data_sources.py` + one field in `CollectorResult` — no other file needs to change. The collector runs 12 workers in parallel.

Lapisan 1 — Makro Global (diambil sekali, bukan per simbol)

Sumber	Penyedia	Auth	Data / Bidang	Digunakan di
VIX	CBOE · yfinance	Free	Current VIX level; 5-min cache	Scanner cap (>35→6.0, >25→7.5) · Confluence ×0.80 when
DXY	ICE · yfinance	Free	USD strength level; regime label	Macro regime block · Scanner Stage 3 header
Fear & Greed	alternative.me	Free	Score 0-100; label (Extreme Fear ...	Extreme Caped · Sentiment prompt · Dashboard pulse
Economic Calendar	finnhub API	Key	FOMC/CPI/NFP events; hours_until; start	Scanner flag when high-impact event in 24h · Prompt r
BTC Dom + Mkt Cap	CoinGecko (free)	Free	btc_dominance_pct; total_market_cap	Scanner market regime · Call Analyzer market context

Lapisan 2 — Struktur Pasar (seluruh kripto, bukan per simbol)

Sumber	Penyedia	Auth	Data / Bidang	Digunakan di
Options Skew (PCR/IV)	Deribit (free)	Free	put_call_ratio; iv_skew; near_term_iv	BTC FV-Score; institutional call/call bias in sentiment
BTC Mempool	blockchain.com	Free	mempool_bytes; n_transactions; avg_fee	One-liner congestion data injected into Call Analyzer
Trending Coins (top 10)	CoinGecko (free)	Free	Top-10 trending coin symbols in last 24h	Are the analyzed coin trending? Injected into prompt con

Lapisan 3 — Tingkat Simbol (per koin yang dianalisis)

Sumber	Penyedia	Auth	Data / Bidang	Digunakan di
Multi-Exchange L/S Ratio	Binance+Bybit+OKX · CCXT	Free	L/S ratio per exchange; consensus diff	On-chain positioning block; on-chain divergence flag (>65% v
OI · Funding · Liquidation	Coinalyze API	Key	Aggregated OI (multi-exchange); 24h	Derivatives block; on-chain divergence flag (>65% v
Cap rank · tier · volume	CoinGecko (free)	Free	market_cap_rank; cap_tier (mega/large	On-chain positioning block; on-chain divergence flag (>65% v
DeFi TVL + 7d change	DefiLlama (free)	Free	protocol; tvl_usd; tvl_7d_change_pct	DeFi block; on-chain divergence flag (>65% v
OHLCV Candles (4H + 1D)	Binance Futures · CCXT	Free	~200-bar OHLCV DataFrame per time	Price action block; S/R detection · SMT divergence · Ba

Lapisan 4 — Kecerdasan Perdagangan (paling spesifik untuk perdagangan yang dianalisis)

Sumber	Penyedia	Auth	Data / Bidang	Digunakan di
Smart Money Wallet Flow	Flowser API	Paid	signal; label; smart_money_bias; accum	Smart money block; on-chain divergence flag (>65% v
Social & News Context	AI Grok API	Key	text (narrative); weight 0.0–0.8 (scaled	Baseline bias block injected into Call Analyzer prompt · lowest prompt bud

5.1 Arsitektur Konteks Grafik

The chart pipeline is split into three pure modules for testability and maintainability:

Modul	Tanggung jawab	Fungsi utama
chart_indicators.py	Pure indicator computation — no A/R calls, no side effects	compute_rsi, compute_macd, compute_ema_alignment, compute_adx, compute_wave_trend
chart_sr.py	S/R detection with ATR-relative tolerance, support/resistance (ATR clustering, exponential decay on touch records)	detect_support_resistance
chart_candles.py	OHLCV fetch + 10-min cache	get_candles (Binance via CCXT), get_candles_for_chart
chart_patterns.py	Trendlines + Fibonacci retracements	detect_trendlines, detect_fibonacci
chart_confluence.py	9-signal confluence scorer + SMT divergence (VIX multiplier $\pm 0.5\%$), _smt_direction_weight (24h correlation)	compute_confluence_score, detect_divergence
chart_context.py	Thin facade — re-exports from all other modules above	get_candles, compute_indicators, confluence_score, get_candles_for_chart

9-signal confluence system: RSI, MACD, EMA, ADX, WaveTrend (VMC Cipher A/B, $n1=10/n2=21$), MFI, CVD (Money Flow Multiplier $v \times (2c-l-h)/(h-l)$), volume anomaly, plus 2 SMT variants. Max score 6.50/timeframe. VIX multiplier applies $\times 0.80$ on the final score when $VIX > 30$ — so macro stress automatically reduces confluence conviction.

6. Agen Otomatisasi

Automation agents run continuously in background threads. They require no user interaction — they watch the market, sync positions, and fire alerts automatically.

6.1 Penjadwal Pemindai — `scanner_scheduler.py`

Runs the full 3-stage scanner pipeline every 30 minutes. First scan fires 5 minutes after app startup (allowing exchange sync to complete). After every run that produces results above the score threshold:

1. Sends a Telegram HTML alert with symbol, direction, score, entry zone, SL, TP1, TP2, R:R, and urgency.
2. Saves each alerted setup to `analyzed_calls` with `analyst='scanner'`. This is critical for automatic position linking — when a scanner-alerted position opens on the exchange, check-matches auto-confirms the link without user action.
3. Deduplicates by (symbol, direction) within a 4-hour window to prevent spam on consecutive scans.

6.2 Sinkronisasi Bursa — `bitget_sync.py` / `blofin_sync.py`

Runs every 5 minutes in a background thread. Uses cursor-based pagination to catch all closed positions regardless of holding duration. Key behaviours:

Fitur	Apa yang dilakukan
Tutup panggilan otomatis	When a position closes, finds the linked <code>analyzed_call</code> and marks it closed. Records which TP or SL was
Penandaan rezim pasar	Tags each position bull/bear/range at entry time using BTC EMA50/200 cross (<code>get_btc_regime</code>). Enables
Pelacakan MFE/MAE	Records Maximum Favourable Excursion and Maximum Adverse Excursion for each trade. Used for the
Deduplikasi	Idempotent — safe to run every 5 minutes. Uses exchange order IDs to prevent duplicate entries regardl
Jendela kejar-ketinggalan	On startup, fetches trades from the last 48 hours to recover any missed during downtime (Pi restart, netw

7. Backtest Tertanam & Optimizer

The embedded backtester runs entirely on historical positions stored in the local SQLite database — no external API calls required. It uses the same indicator logic as the live pipeline (WaveTrend $n1=10/n2=21$, CVD MFM formula) so backtest results are comparable to live signal quality.

7.1 Mesin Backtest — `backtest_engine.py`

Komponen	Apa yang dilakukan
<code>run_backtest(symbol, tf, days, parameters, offset_days)</code>	Fetches OHLCV data, runs CXT, applies vectorised signal logic, simulates trades, returns <code>BacktestResult</code>
<code>backtest_metrics.py</code>	Pure metric functions: <code>sharpe_ratio</code> (sample std $N-1$, annualised $\sqrt{365}$), <code>sortino_ratio</code> (downside-only)
Configurable params	<code>rsi_oversold</code> (25–45), <code>rsi_overbought</code> (55–75), <code>ema_short/long</code> (10–50/50–250), <code>adx_min</code> (15–35), a

7.2 Optimizer Bayesian — `backtest_optimizer.py`

Uses Optuna (TPE sampler) to maximise Sharpe ratio over 7 parameters. Runs in a background daemon thread so the UI stays responsive. Each run is stored in the `optimizer_runs` table — the Analysis tab shows the last 5 runs with Sharpe, win rate, and best parameters.

7.3 Uji Walk-Forward — Tanpa Kebocoran Data

The walk-forward test splits the real position date range 70% training / 30% test. The critical implementation detail: `end_offset_days` is threaded through `run_backtest` → `_fetch_ohlc` so the training window ends at `now – test_days` (not at `now`). Without this, both windows anchor to the present — the test set is a subset of the training set, making all walk-forward results meaningless.

Jendela	Rentang pengambilan	Sumber parameter	Tujuan
Pelatihan (70%)	<code>now – (test+train days) → now – test_days</code>	Optimizer maximises Sharpe	Pencarian parameter terbaik
Uji (30%)	<code>now – test_days → now</code>	Training best params applied	Evaluasi Sharpe di luar sampel
Sinyal overfitting	<code>train_sharpe >> test_sharpe</code>	—	Selisih > 0.5 menunjukkan curve-fitting; gunakan para

7.4 Metrik Dasbor — `analytics.py`

Sharpe and Calmar are computed from `wallet_snapshots` (rolling balance history imported from exchange). Key formula invariants:

Metrik	Rumus	Catatan
Rasio Sharpe	$\text{mean}(\text{daily_ret}) \times 365 / (\text{std}(\text{daily_ret}, N-1) \times \sqrt{365})$	Varians sampel (penyebut $N-1$). Filter dompet: saldo > \$1 USDT.
Rasio Calmar	$\text{ann_return_pct} / \text{max_dd_pct}$	<code>max_dd_pct</code> diukur sebagai % dari puncak bergulir SAAT LEMBA
Volatilitas tahunan	$\text{std}(\text{daily_ret}, N-1) \times \sqrt{365} \times 100$	Ditampilkan sebagai % di bawah Sharpe pada dasbor.

8. Arsitektur Prompt & Caching

How prompts are built and cached is one of the most important architectural decisions in the system — it directly affects both cost and accuracy.

8.1 Pemisahan Stabil / Dinamis

Anthropic's prompt caching works by storing a prefix that is byte-for-byte identical across calls. When the cached prefix is reused, Anthropic charges approximately 10% of the normal input token price for the cached portion. However, if any live data (funding rates, chart indicators, market context) is included in the cached block, the cache key changes every few minutes and cache hits never occur — you pay full price on every call.

The fix: **build_stable_prefix()** returns only content that changes at most weekly (rulebook + calibration + pattern strengths). **build_context()** returns dynamic content that changes per call (backtest insights, market data, chart indicators, Nansen, Grok, similar trades). The stable prefix gets `cache_control: ephemeral` — the dynamic context does not.

Blok	Isi	Cache?	Berubah seberapa sering
Prefiks stabil (Blok 1)	Rulebook (10 rules), calibration feedback, top-3 pattern strengths	✓ YES <code>cache_control: ephemeral</code>	Mingguan (saat 5+ perdagangan baru)
Konteks dinamis (Blok 2)	Backtest insights, live market context, chart indicators, Nansen, Signal, Grok, data (bait, paris, set, 5 cal)	✗ NO	Setiap 5 menit
Prompt variabel (Blok 3)	Call text, position sizing, CoT from previous same symbol analysis	✗ NO	Every call

8.2 Loop Umpan Balik Backtest

Every Claude analysis prompt begins with a compact historical performance block injected by **get_backtest_context()** in `analytics.py`. This gives Claude specific, numerical context about YOUR trading patterns before it scores any new call:

Contoh konteks backtest yang disuntikkan ke prompt:

BACKTEST INSIGHTS: Recent form: 72% WR last 20 · streak WWLWW · avg +\$8.40 Breakout setups: 100% WR (6 trades) avg +\$7.00 BTCUSD Long: 75% WR (12 trades) avg +\$12.50 ■ Wednesday: caution (57% WR, -\$355 total P&L;) ■ 21:00 UTC: weak hour (70% WR, -\$1831 total)

This is not generic advice — it is derived from the user's actual trade history in real time. Claude sees both the opportunity signal (technical setup) and the historical context (does this trader actually profit from this type of setup at this time of day?) before assigning a score. This is the primary mechanism for improving accuracy as trade history grows.

8.3 Loop Pembelajaran CoT

When Claude analyzes a call, its step-by-step reasoning (the 'thinking' field in the JSON response) is stored as `cot_reasoning` in the database. The next time the same symbol is analyzed, the previous reasoning is injected into the prompt as `PREVIOUS ANALYSIS` context. Claude can then explicitly compare: 'Last time I analyzed ARKMUSD, I flagged the SL as too close to the 4H noise floor. Has that changed?' This enables detection of repeated mistakes and continuous refinement without any retraining.

9. Sistem Tautan Posisi Otomatis

A key feature is that scanner-alerted trades and manually analyzed calls are automatically linked to the corresponding live positions — without requiring the user to confirm each match manually.

9.1 Cara tautan dibuat

Skenario	Perilaku tautan otomatis
Scanner sends Telegram alert for NOTUSDT Long entry .	<code>scanner._persist_setups()</code> saves the setup to <code>analyzed_calls</code> (analyst='scanner', status='open').
User ran call analyzer for NOTUSDT Long, position closed, when the position closed.	When a new NOTUSDT Long opens, check-matching logic will find the previous call and link it to the new position.
Trade came from Telegram but not call live position analyzed (WSXUSDT call).	'Analyze First' button. Clicking it navigates to the call analyzer.
Scanner signal didn't save (scored below threshold) but position was opened.	Position is created directly from the live position data with analyst='scanner'. check-matching logic will find the previous call and link it to the new position.

9.2 Yang muncul di Perdagangan Live saat ditautkan

Once a position is linked to a call, the live trades card shows a **Call Targets Panel** with: distance from mark price to SL, TP1, TP2, and the call's average entry. A TP1-reached alert fires when mark price crosses TP1, with an automatic break-even stop suggestion. Positions with a linked call also show the setup score, trade type, and R:R ratio from the original analysis.

10. Pengukuran Akurasi & Target 85%

The accuracy target of $\geq 85\%$ means: when the consensus score is ≥ 6 and confidence is 'high' (both Claude and Gemini agree within 1 point), the trade should be profitable at least 85% of the time. This is measured by `scripts/backtest_consensus.py`.

10.1 Tiga hipotesis yang diuji

Hipotesis	Klaim	Cara diukur
H1: Claude saja	Skor ≥ 6 dari Claude saja memprediksi pergerakan harga	pergerakan harga di masa datang dengan menggunakan <code>get_calls</code> with <code>setup_score $\geq N$</code>
H2: Konsensus	Kesepakatan antara Claude dan Gemini (pilih) meningkatkan akurasi	(pilih) meningkatkan akurasi dengan <code>consensus_score $\geq N$ AND confidence='high'</code>
H3: Penghindaran Pengulangan	Pengulangan dengan $ \Delta > 2$ (tanda TINJAU)	metode ini akan bekerja dengan baik jika <code>abs(consensus_score - 6) > 2</code>

Current status: 5 outcome-recorded calls (need ≥ 20 for statistical confidence). The system is now accumulating evidence — every new call saved stores `gemini_score` and `consensus_score`, and every outcome recorded improves the backtest context injected into future prompts. The 85% target becomes measurable after ~15-20 more outcome-recorded calls.

Run the backtest at any time: `python3 scripts/backtest_consensus.py --host <pi-ip>:8082`. Add `--live` to re-score the last 20 calls with Gemini live (uses API credits).

11. Pipeline Agen Khusus (v1.5.0)

In v1.1.0 the AI pipeline was refactored into 7 specialized agents with typed input/output contracts (TypedDict). Each agent has one clear responsibility, can be tested in isolation, and communicates only via return values — no shared mutable state. All TypedDicts live in agent_types.py.

11.1 Alur Pipeline

DataCollector → [DataInterpreter + MarketSentiment (parallel)] → DataReviewer → TradePrep (Claude + Gemini) → RiskMgmt → AnalysisResult
After position opens: TradeMonitor (background, every 10 min)
runs: DataCollector → DataInterpreter → MarketSentiment → Haiku verdict
On risk_rating ≥ 7 or action ≠ Hold: fires Telegram alert + sets UI badge

11.2 Kontrak Agen

Agen	Masukan	Keluaran	Panggilan AI?	Akses DB?
DataCollector	CollectorInput	CollectorResult	No	No
DataInterpreter	CollectorResult	InterpreterResult	No	No
MarketSentiment	CollectorResult	SentimentResult	No	No
DataReviewer	InterpreterResult	ReviewerResult	No	Hanya baca
RiskManagement	TradePrepResult	RiskResult	No	No
TradePreparation	All 4 above	TradePrepResult	Sonnet+Gemini	Baca (prefiks stabil)
TradeMonitor	Position + Interp	MonitorResult	Haiku	Read
ChartDraw	Candles + levels	PNG (base64)	No	No

11.3 Kemampuan Baru

■ Annotated Trade Charts — agent_chart_draw.py

v1.1.0

Trigger: Otomatis — dipicu oleh TradePrep atau thread monitor

Output: See description

When TradePrep produces a trade recommendation, agent_chart_draw.py generates a dark-themed mplfinance candlestick chart annotated with Entry (blue dashed), Stop Loss (red dashed), TP1 and TP2 (green) lines, a level legend, and the decision criteria as text overlaid in the top-right. The PNG is base64-encoded and stored in analyzed_calls.chart_png_b64. Telegram scanner alerts attach this chart as a photo — you see the trade setup visually, not just as numbers.

■ Kelly Criterion Sizing — agent_risk_mgmt.py

v1.1.0

Trigger: Otomatis — dipicu oleh TradePrep atau thread monitor

Output: See description

Position sizing now includes a Kelly criterion fraction (0.05–0.25) derived from the setup_score as an edge proxy. Kelly maps a score of 1-10 to a win-rate estimate of 0.35–0.75, then computes $f = (WR \times R - (1 - WR)) / R$ where $R = 2.0$ (conservative 2:1 R:R baseline). Capped at 0.25 to prevent overbetting. The sizing_breakdown and kelly_fraction are saved in analyzed_calls.risk_verdict_json for review.

■ Proactive Position Monitor — monitor_scheduler.py

v1.1.0

Trigger: Otomatis — dipicu oleh TradePrep atau thread monitor

Output: See description

A background thread polls all open positions every 10 minutes. Positions where unrealized_pct < -5% OR duration > 4 hours are checked with a lightweight DataCollector → DataInterpreter → MarketSentiment → Haiku chain. When risk_rating ≥ 7 or action ≠ Hold, the system fires a Telegram alert and sets monitor_alert=1 in analyzed_calls for the UI badge — without executing any trades (recommend-only).

■ Contra Signal Detection — agent_market_sentiment.py

v1.1.0

Trigger: Otomatis — dipicu oleh TradePrep atau thread monitor

Output: See description

The MarketSentiment agent computes a contra_signal flag: True when the crowd is heavily positioned against the proposed trade direction (>65% of accounts long when you're going Long). Contrarian awareness is injected into every TradePrep prompt, and the TradeMonitor uses it to raise risk ratings on existing positions that are swimming against the crowd.

12. Referensi Agen Lengkap

Agen / Modul	Jenis	Model	Pemicu	Anggaran token
call_analyzer	Analisis	Sonnet 4.6	Sesuai permintaan	~4,000 in / 4,096 out
scanner (batch)	Analisis	Sonnet 4.6	Otomatis 30 menit	~5,500 in / 14,400 out
scanner (quick)	Analisis	Haiku 4.5	Per finalis	~1,200 in / 120 out
advisor	Analisis	Sonnet 4.6	Sesuai permintaan	~4,000 in / 4,096 out
rulebook	Analisis	Sonnet 4.6	Mingguan / manual	~3,000 in / 2,048 out
hindsight	Analisis	Haiku 4.5	Batch sesuai permintaan	~800 in / 512 out
live_trade	Analisis	Haiku 4.5	Per klik	~600 in / 768 out
trade_grader	Analisis	Haiku 4.5	Per perdagangan ditutup	~700 in / 350 out
limit_analyzer	Analisis	Sonnet 4.6	Per order limit	~2,000 in / 768 out
pattern_detector	Analisis	Sonnet 4.6	Melalui advisor	~2,500 in / 1,200 out
Gemini 2.0 Flash	AI Eksternal	Gemini 2.0	Paralel dengan panggilan	~900 in / 200 out
xAI Grok 3 Fast	AI Eksternal	Grok 3	Paralel dengan panggilan	~950 in / 130 out
Nansen screener	Data	—	Per pemindaian / panggilan	1x per credit per run
chart_context	Data	—	Per analisis	Bitget REST (ter-cache)
market_context	Data	—	Per analisis	4 bursa + 2 API
scanner_scheduler	Otomatisasi	—	Setiap 30 menit	Menjalankan pemindai + Telegram
monitor_scheduler	Otomatisasi	—	Setiap 10 menit	Haiku per posisi
bitget_sync	Otomatisasi	—	Setiap 5 menit	Kursor Bitget REST
blofin_sync	Otomatisasi	—	Setiap 5 menit	Blofin REST cursor
agent_data_collector	Agent	—	Per panggilan pipeline	Paralel: 12 sumber (4 lapisan)
agent_data_interpreter	Agent	—	Per panggilan pipeline	Murni: indikator
agent_market_sentiment	Agent	—	Per panggilan pipeline	Murni: bias makro
agent_data_reviewer	Agent	—	Per panggilan pipeline	Baca DB: KPI
agent_risk_mgmt	Agent	—	Per panggilan pipeline	Matematika murni: Kelly
agent_trade_prep	Agent	Sonnet+Gemini	Per panggilan pipeline	Panggilan AI utama
agent_trade_monitor	Agent	Haiku 4.5	Per pass monitor	~800 in / 300 out
agent_chart_draw	Agent	—	Per TradePrep	PNG mplfinance